Supermarket Sales Dashboard Project Report

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1. Introduction

In modern business environments, data plays a crucial role in decision-making. Retail businesses generate large amounts of transactional data every day, but without proper analysis, this data remains unused.

This project focuses on analyzing supermarket sales data using SQL and Power BI to transform raw transactional records into meaningful business insights. The final output is an interactive dashboard that allows stakeholders to understand sales performance, customer behavior, and product trends in a clear and visual format.

2. Project Overview

This project follows a complete Business Intelligence (BI) workflow, starting from raw data extraction to dashboard visualization.

The main stages of the project include:

- Data extraction from a MySQL database (XAMPP environment)
- Data cleaning and preprocessing
- SQL-based analytical queries
- Dashboard creation using Power BI
- Insight generation for business decision-making

The final result is a structured and interactive dashboard that supports strategic planning in retail operations.

3. Objective

The main objectives of this project are:

- To analyze supermarket sales performance across different branches and time periods
- To identify the best-performing products and categories
- To understand customer purchasing behavior and preferences
- To evaluate payment method trends
- To develop an interactive Power BI dashboard for business insights and reporting

4. Dataset Description

Source

The dataset used in this project is a supermarket sales dataset stored in a MySQL database (XAMPP environment). It contains real-world retail transaction data.

Number of Records

The dataset contains approximately 1000+ transaction records, depending on the version used.

Features

The dataset includes the following attributes:

- Invoice ID – Unique transaction identifier
- Branch – Store branch (A, B, C)
- City – Location of the branch
- Customer Type – Member or Normal customer
- Gender – Male or Female
- Product Line – Category of purchased products
- Unit Price – Price per item
- Quantity – Number of items purchased
- Tax / Gross Income – Tax applied on purchases
- Total Sales – Final transaction value
- Date – Transaction date
- Payment Method – Cash, E-wallet, or Card
- Rating – Customer satisfaction score

5. Data Cleaning

Before analysis, the dataset was cleaned and transformed to ensure accuracy and consistency.

Data Cleaning Steps

- Removed missing or null values to ensure data completeness
- Fixed inconsistent or incorrect data entries
- Standardized categorical values (e.g., product names, payment methods)
- Corrected data types (numeric, text, and date formats)

Added Calculated Columns

To enhance analysis, additional columns were created:

- **Month Column** → Extracted from the Date field to analyze monthly trends
- **Day Name Column** → Extracted from Date to identify weekday patterns

These transformations made it easier to perform time-based analysis in Power BI.

6. SQL Analysis

SQL queries were used to perform structured data analysis and generate business insights.

Monthly Sales Analysis

Monthly sales trends were analyzed to identify seasonal patterns and peak sales periods. This helped in understanding how sales fluctuate over time.

Branch Analysis

Each branch (A, B, and C) was analyzed based on total revenue and performance. This comparison helped identify the most successful branch in terms of sales.

Product Analysis

Product categories were evaluated to determine:

- Top-selling product lines
- Revenue contribution per category
- Product performance trends

This analysis provided insights into which products generate the highest business value.

7. Power BI Dashboard

An interactive Power BI dashboard was developed to visualize the analyzed data.

Supermarket Sales Dashboard

This dashboard provides a high-level summary of business performance, including:

- Total Revenue (KPI Card)
- Total Profit (KPI Card)

- Total Transactions (KPI Card)
- Average Customer Rating (KPI Card)
- Monthly Sales Trend (Line Chart)

This page is designed for executives to get a quick overview of business health.

Product Performance Dashboard

This dashboard focuses on product-level insights:

- Sales by Product Category
- Sales Quantity Analysis
- Product Profit Analysis
- Average Rating Analysis

It helps identify which product categories perform best in the market.

Customer Insights Dashboard

This section analyzes customer behavior:

- Sales Performance by Gender
- Customer contribution to sales
- Sales by Payment method
- Customer rating behavior

This helps businesses understand customer preferences and improve services.

8. Business Insights

From the analysis, several important business insights were identified:

- Electronic accessories and food-related products performed strongly in terms of sales
- Branch A consistently generated the highest revenue compared to other branches
- E-wallet and cash were the most commonly used payment methods
- Female customers tended to give slightly higher ratings than male customers
- Sales showed seasonal variations, indicating demand changes over time

These insights can help businesses optimize marketing strategies and improve operational decisions.

9. Key Findings

The key findings from this project include:

- Certain product categories dominate sales and should be prioritized in inventory planning
- Customer behavior varies significantly based on gender and customer type
- Branch performance is not evenly distributed, indicating the need for performance optimization
- Digital payment methods are becoming increasingly popular among customers
- Time-based patterns can be used to predict future sales trends

10. Conclusion

This project demonstrates how raw supermarket transaction data can be transformed into meaningful business intelligence using SQL and Power BI.

By combining data cleaning, structured SQL analysis, and interactive dashboard visualization, the project provides valuable insights that support better decision-making.

Overall, this system helps businesses:

- Understand sales performance
- Identify profitable products
- Analyze customer behavior
- Improve strategic planning

This makes the project a complete end-to-end Business Intelligence solution for retail analytics.